AN INFINITE FAMILY OF COUNTEREXAMPLES TO AN ATOMIC-WEIGHT CONJECTURE OF DARGAD AND LARSSON

Abstract. Dargad and Larsson conjectured an atomic-weight formula for truncated-support partizan subtraction games. We prove that, for every integer $t \geq 1$,

$$\mathcal{T}_t(2t+4; 1, 2t+1) = \downarrow.$$

Its atomic weight is -1, whereas Conjecture 7.1 in version 1 of their preprint predicts 0. In these instances $a = 1$, and the formula fails already at the smallest heap size it governs. The equality $\tau = (b-a)/2$, which these instances also satisfy, is not itself the obstruction: the formula does hold at that boundary in instances tabulated by Dargad and Larsson. Their theorem for $\tau < (b-a)/2$ is unaffected.

1. The counterexamples

In the truncated-support game $\mathcal{T}_\tau(n; a, b)$, Left may subtract any integer in $[a] = \{1, \ldots, a\}$ from a heap of size n, while Right may subtract any integer in $\{\tau+1, \ldots, b\}$ [2, Definition 5.1]. Play is normal: a player with no legal move loses [1]. In version 1 of their preprint, Dargad and Larsson conjectured that, for $a < b$, $\tau < b - a$, and $n \geq R_{\widehat{\beta}}$,

$$\operatorname{aw}\big(\mathcal{T}_\tau(n; a, b)\big) = -\left\lfloor \frac{n - R_{\widehat{\beta}}}{a} \right\rfloor, \tag{1}$$

where

$$\widehat{\beta} = \left\lfloor \frac{b-a}{b-a-\tau} \right\rfloor, \qquad R_{\widehat{\beta}} = \widehat{\beta}(a + \tau + 1)$$

[2, Equations (5), (13), and Conjecture 7.1].

We use the standard notation

$$* = \{0 \mid 0\}, \qquad \uparrow = \{0 \mid *\}, \qquad \downarrow = \{* \mid 0\} = -\uparrow.$$

We also use the standard fact that a short game equals 0 precisely when the second player wins, whether the second player is Left or Right [1].

Theorem 1. *Fix an integer $t \geq 1$ and write*

$$H_n = \mathcal{T}_t(n; 1, 2t+1).$$

Then

$$H_{2t+2} = 0, \qquad H_{2t+3} = *, \qquad H_{2t+4} = \downarrow.$$

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Proof. Here Left removes one token, while Right removes any number from $t+1$ through $2t+1$.

First, $H_{2t+2} = 0$. If Left starts, she leaves $2t+1$ tokens and Right removes the entire heap. If Right starts, he leaves a heap of size $m \in \{1, \ldots, t+1\}$; Left moves to $m-1 \le t$, after which Right has no legal move.

Next consider $H_{2t+3} + *$. If Left first moves in the heap, Right moves $* \to 0$, leaving $H_{2t+2} = 0$. If Left instead moves $* \to 0$, Right reduces the heap to $t+2$, after which the heap moves

$$t+2 \xrightarrow{\mathrm{L}} t+1 \xrightarrow{\mathrm{R}} 0$$

finish the game.

Suppose instead that Right starts. If he moves $* \to 0$, Left reduces the heap to $2t+2$. If he moves in the heap, he leaves $m \in \{2, \ldots, t+2\}$, and Left moves to $k = m-1 \in \{1, \ldots, t+1\}$. A subsequent move $* \to 0$ by Right is answered by $k \to k-1 \le t$, leaving Right without a legal move. A subsequent heap move by Right is possible only when $k = t+1$, and Left then moves $* \to 0$. Thus $H_{2t+3} + * = 0$, so $H_{2t+3} = *$.

Finally consider $H_{2t+4} + \uparrow$. If Left first moves in the heap, Right moves $\uparrow \to *$, leaving $H_{2t+3} + * = 0$. If Left instead moves $\uparrow \to 0$, Right reduces the heap to $t+2$, and the same two heap moves as above finish the game.

Suppose that Right starts. If he moves $\uparrow \to *$, Left reduces the heap to $2t+3$, again leaving $H_{2t+3} + * = 0$. If he moves in the heap, he leaves $m \in \{3, \ldots, t+3\}$, and Left moves to $k = m-1 \in \{2, \ldots, t+2\}$. If Right next moves in the heap, at most one token remains; Left moves $\uparrow \to 0$, and Right has no legal move. If Right instead moves $\uparrow \to *$, Left reduces the heap to $k-1 \le t+1$. A move $* \to 0$ by Right is then answered by one further Left heap move, leaving at most t tokens. The only alternative is a Right heap move, which is possible only from a heap of size $t+1$ and is answered by $* \to 0$. Hence $H_{2t+4} + \uparrow = 0$, and therefore $H_{2t+4} = -\uparrow = \downarrow$. □

Corollary 2. *Conjecture 7.1 in version 1 of Dargad and Larsson* [2] *is false as stated.*

Proof. For any $t \ge 1$, take

$$(a, b, \tau, n) = (1, 2t+1, t, 2t+4).$$

Then $\tau = t < 2t = b - a$, and

$$\widehat{\beta} = \left\lfloor \frac{2t}{t} \right\rfloor = 2, \qquad R_{\widehat{\beta}} = 2(1 + t + 1) = 2t + 4 = n.$$

Formula (1) therefore predicts atomic weight 0. By Theorem 1, however, the game is $\downarrow$. Since $\operatorname{aw}(\uparrow) = 1$ and $\operatorname{aw}(-G) = -\operatorname{aw}(G)$,

$$\operatorname{aw}(\downarrow) = -1$$

[2, Definition A.17 and Lemma A.19(1)]. This is a contradiction. □

Remark. Every instance above has $a = 1$ and $\widehat{\beta} = 2$, so the failure occurs at $n = R_{\widehat{\beta}}$, the smallest heap size to which (1) applies. These instances also satisfy $\tau = (b-a)/2$, but that equality is not itself the obstruction: for $(a, b) = (3, 7)$ and $\tau = 2 = (b-a)/2$ one has $\widehat{\beta} = 2$ and $R_{\widehat{\beta}} = 12$, and the atomic weights tabulated in [2, Table 4] agree with (1) for every tabulated n with $12 \leq n \leq 34$. The theorem proved in [2, Theorem 1.5] assumes the strict inequality $1 \leq \tau < (b-a)/2$ and is therefore unaffected. We do not determine here how far beyond the family of Theorem 1 the failure of (1) extends.

References

[1] M. H. Albert, R. J. Nowakowski, and D. Wolfe, *Lessons in Play: An Introduction to Combinatorial Game Theory*, 2nd ed., A K Peters/CRC Press, 2019.

[2] A. Dargad and U. Larsson, *Partizan Subtraction with Full and Truncated Support*, arXiv:2607.27989v1 [math.CO], 2026.